Secure Network Merger Plan: Company A & Company B

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When merging networks, they are considered challenging because two companies have different systems and security requirements. Company A is a financial services company, and Company B specializes in medical software. This report will outline the problems in each company's network; they will both identify risks based on vulnerability scans and propose a new network design that enables both companies to continue operating while the merger is in progress. The design will incorporate **Zero Trust, cloud support, and strong security tools, all while staying under the $50,000 budget.**

A. Current Network Security and Infrastructure Problems

Company A and Company B both have weaknesses in their security and technology. These problems make them target hackers and slow down employees who are trying to work.

Company A (Financial Services)

Security Problems:

- Employees can be tricked by phishing emails that steal passwords.
- Hackers could use ransomware to lock financial data.

Infrastructure Problems:

- Older computers slow down work and software.
- Banking rules change often and require constant updates.

Company B (Medical Software)

Security Problems:

- Some medical devices do not get security updates fast enough.
- Vendors have access to internal systems, which increases risk.

Infrastructure Problems:

- Older systems cause slowness and sometimes freezing.
- The network slows down when many users are connected.

B. Analysis of Network Diagrams and Vulnerability Scans

Both companies have weaknesses found within their network setup.

B1. Existing Vulnerabilities:

Company	Vulnerability	Description
A	Outdated servers	Older software has known security flaws

A	Weak remote login	Remote access does not always require MFA
B	Default passwords	Devices still use factory passwords
B	Vendor access	Vendors login without strong checks

B2. Impact, Risk, and Likelihood:

Based on NIST SP 800-30.

Issue	**Impact**	**Risk**	**Likelihood**
A-Outdated Servers	High	High	Medium
A- Weak Remote Login	Medium	High	High
B-Default Passwords	High	Critical	High
B-vendor Access	Medium	High	Medium

A- Outdated Servers- Outdated servers **Impact** is very high due to older software systems being known for having flaws that hackers/attackers can easily access. When someone takes advantage of the flaws, they are able to exploit and change financial records. This hurts two

of the CIA triads which are confidentiality and integrity. The **risk** is high because Company A is a financial business that handles sensitive money data, so attackers love making business such big targets. The **likelihood** is medium because once a weak server is found it can be attacked, but not every system is exposed.

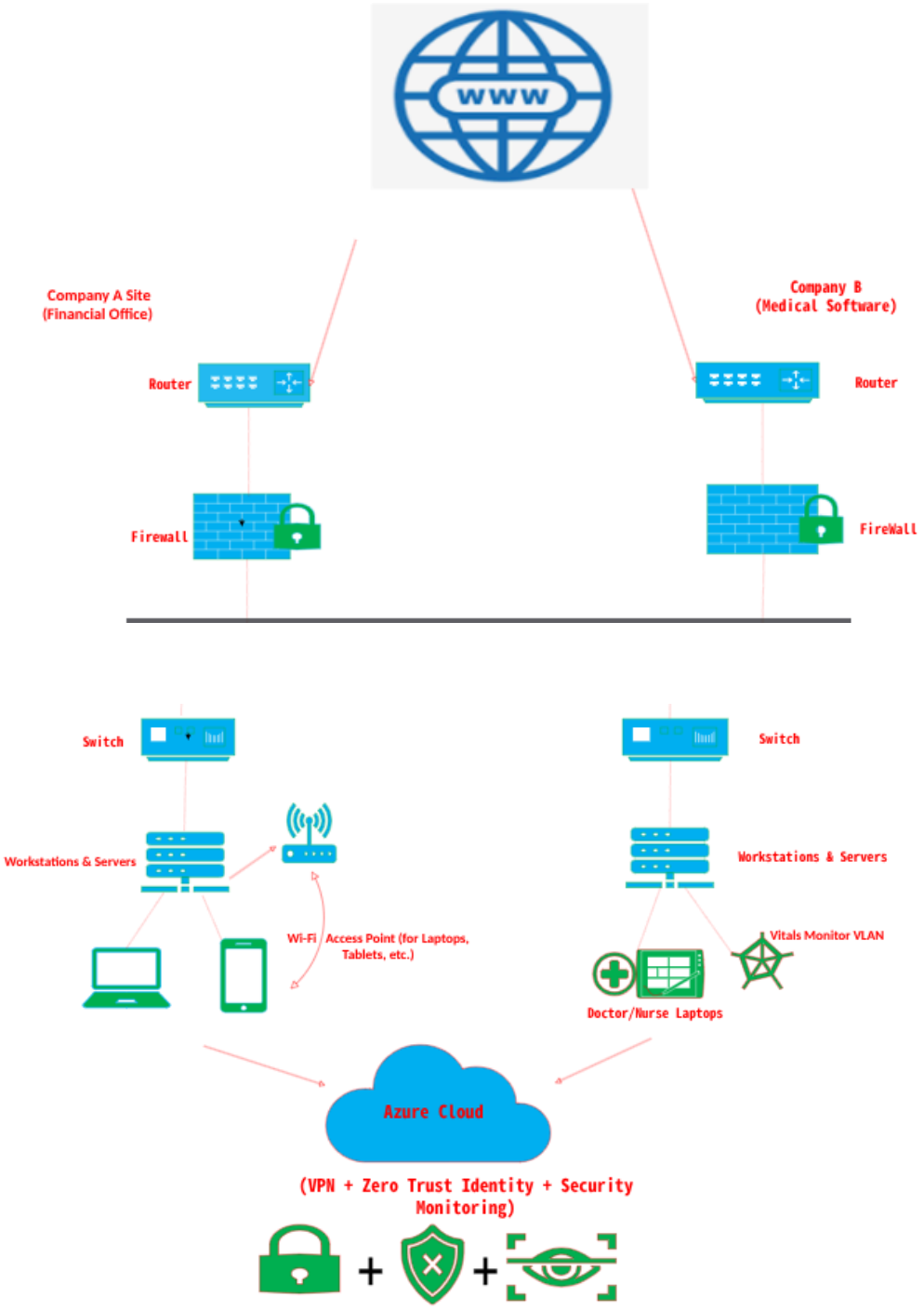
A- Weak Remote Login- The **impact** of a weak remote login is medium because attackers who can steal passwords can log in from outside of a company and view internal systems. The **risk** is High because remote access is reachable from the internet which is a common target. The **likelihood** is high because password guessing and phishing are very common and without strong protection like an MFA, things like stolen customer data and setting changes will often work.

B-Default Passwords- The **impact** of a default password is high because devices with factory passwords are easy to break into. Hackers are very aware of how easy that can be. Once they have access, they change settings and move deeper in the network or exploit patient information, which is a violation of the HIPPA law. The **risk** is critical because the weakness gives hacker control over devices that can be connected to other systems. The **likelihood** is high because many attackers automatically scan for devices still using default credentials

B-vendor Access- The **impact** of vendor access is medium. A vendor can access internal systems to provide support. When a vendor account is misused, an attacker logs in as a trusted partner and navigates within Company B's network. The **risk** is high when vendors lack strong security practices, and Company B relies on them to support its medical software. The **likelihood** is medium because not every vendor will be attacked; however, it only takes one vendor to become compromised. Now, the attacker can use that access to get into Company B’s systems.

C. Proposed Merged Network Topology

Both companies stay in their own buildings with their own routers, firewalls, switches, and workstations. They connect securely using Microsoft Azure cloud. This lets both companies keep working normally during the merger. Zero Trust settings require every user to verify their identity each time they connect.



D. Topology Component Layers

This table shows the OSI and TCP/IP layers for each major component in the merged network:

Device	**Highest OSI Layer**	**Highest TCP/IP Layer**
Router	Layer: 1 (Physical), Layer 2 (Data Link), Layer 3 (Network)	Internet, Network Access
Firewall	Layer: 3 (Network), Layer 4 (Transport), Layer 7 (Application)	Internet, Transport, Application
Switch	Layer 1 (Physical), Layer 2 (Data Link)	Network Access
Workstations & Servers	Layer 1-7 (Physical through Application)	Network Access, Internet, Transport, Application
Azure VPN & Security Tools	Layers 3, 4, and 7	Internet, Transport, Application

E. Rationale and Budget

To stay under the $50,000 budget, we reuse equipment already owned by both companies. We only add cloud tools and security features that help the merger and improve protection.

Item	**Action**	**Cost**	**Reason**

Azure Cloud Security Services	Add	$10,000 per year	Shared secure connection backbone
MFA (Zero Trust Login)	Add	$5,000 per year	Stops stolen passwords from working
Existing routers, switches, and firewalls	Reuse	$0	Still work and support updates
Vendor access control rules	Improve	$0	Fix through settings, not new hardware

F. SECURE NETWORK DESIGN PRINCIPLES

The merged network uses two secure network design principles that connect to the new topology diagram. The principles being presented have made the security of both Company A and Company B stronger and has combined their environments and begun using shared cloud resources.

1. **Network Segmentation and Zero Trust**

The merged design keeps Company A's network and Company B's network in separate segments while they are connected through the cloud. Zero Trust settings in Azure require MFA requirements, and firewall filtering shown in the diagram ensures that users must prove their identity each time they connect, and traffic between segments is checked instead of automatically trusted. This stops attackers from moving freely between the two companies if they break into one part of the network.

2. Cloud Security Controls (Azure Security Tools)

Azure cloud uses built-in tools to protect, encrypt connections, identity management, and traffic inspection. The topology diagram shows the Azure VPN gateway and cloud security services acting as the central connection point between Company A and Company B. The tools will be able to monitor traffic, enforce conditional access, and secure data as it travels to and from the cloud. This aligns with the design because the cloud becomes the secure bridge between the two companies rather than a direct network-to-network connection.

G. Regulatory Compliance

The merged network must always follow laws that protect financial and medical data.

Regulation	Why does it Applies	How does the Design Meets It
PCI DSS	Company A mainly handles financial and cardholder data	Firewalls, encryption, and secure cloud storage protect money information
HIPAA	Company B mainly works on patient-related information	Distributor access limits and MFA protect sensitive health data

H. Emerging Threats

Threat	Risk	Performance Impact	Fix
Phishing attacks can accrue	More workers mean more targets due to workers becoming a victim	Time that is lost because of password changings	Multi-Factor Authentication (MFA) + ongoing worker training

Cloud setup mistakes	Could expose data if configured wrong	Slight slowdown from monitoring tools	Azure audits + automated checks

I. Summary and Recommendation

This network design allows both companies to stay separate while still being connected through the cloud. The plan as corrected was to have it budget-friendly and ready to protect against common attacks, such as phishing, while also meeting major laws which include PCI DSS and HIPAA. Using Microsoft Azure enhances security and provides companies with the flexibility to grow in the future. Zero Trust ensures that only authorized employees and approved vendors can access data. This design is the best option because it enables both companies to run smoothly and safely during the merger.

J. References

NIST. (2012). Guide for conducting risk assessments (SP 800-30 Rev. 1). National Institute of Standards and Technology.

Class Scenario. (2025). Company A network of notes and problem descriptions (Course-provided material).

Class Scenario. (2025). Company B vulnerability notes and network documentation (Course provided material).

Microsoft Azure. (2025). VPN gateway and Zero Trust security tools. https://azure.microsoft.com/

K. Professional Communication

This report shows technical information using clear and simple language, while showing a strong understanding of network security. It is organized with proper APA formatting to support easy reading and review by professional audiences.